

PROJECT AND RESOURCE MANAGEMENT SYSTEM

SOLUTION ARCHITECTURE

USER LAYER



Student

- Create Project
- Request Resources
- Track Progress



Technical Faculty

- Review Proposals
- Monitor Projects
- Respond Tickets



Lab Technician

- Manage Inventory
- Process Requests
- Issue & Return



Senior Faculty

- Approve Proposals
- Review Progress
- Generate Reports



Admin

- Manage Users
- Configure System
- View Analytics

ACCESS LAYER



API GATEWAY

Authentication | Authorization | Rate Limiting | Load Balancing | SSL Termination

BUSINESS SERVICE LAYER (MICROSERVICES)

1



User & Access Management Service

- User Registration
- Login / Logout
- Role-Based Access
- Profile Management
- Password Reset
- Account Recovery
- User Activity Management

2



Team & Faculty Assignment Service

- Create / Manage Teams
- View Available Teams
- Self Assign (Faculty)
- View Assigned Faculty
- Team Details Management

3



Project & Proposal Management Service

- Submit Problem Statement
- Add Objectives, Budget & Requirements
- Track Status
- Approve / Reject / Feedback

4



Resource & Inventory Management Service

- Add / Update / Delete Resources
- Track Stock & Availability
- Categorization
- Low Stock Alerts

5



Request & Booking Management Service

- Component Requests
- Update / Cancel Requests
- Approve / Reject Requests
- Equipment Booking & Slot Reservation

6



Issue, Return & Damage Management Service

- Issue Resources
- Return Processing
- Overdue Tracking
- Damage / Loss Management
- History & Penalties

7

Project Progress Tracking Service

- Update Milestones
- Monitor Assigned Teams
- Monitor Overall Progress
- Progress Analytics

8

Document Management Service

- Upload Documents
- Review / Approve / Reject / Comment
- Resubmit Documents
- Version Management

9



Ticket Management Service

- Raise Tickets
- Respond to Tickets
- Track Ticket Status
- Ticket History

10

Notification & Alert Service

- Approval / Rejection Alerts
- Request Updates
- Return Reminders
- Low Stock Alerts
- Notification Inbox

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Knowledge Base / Repository Service

- Store Reports & Logs
- Browse Previous Projects
- Search Repository Content

12

Reporting & Analytics Service

- Generate Reports (Projects, Inventory, Tickets)
- Download Reports
- Dashboards & Analytics

DATA LAYER



User Database



Project Database



Inventory Database



Request Database



Booking Database



Document Storage



Log & Audit Database



Notification Database



Analytics Database

INFRASTRUCTURE LAYER



Cloud / On-Premise Infrastructure



Application Server (Scaling Group)



Database Server (Primary)



File Storage (Cloud / NAS)



Backup & Disaster Recovery



Monitoring & Logging (ELK / Prometheus)

ARCHITECTURE OVERVIEW

The Project and Resource Management System is a multi-tier web application that enables efficient management of projects, resources, users, and workflows across the organization. It ensures transparency, real-time tracking, and role-based access control.

KEY FEATURES

- ✓ Advanced Security & Role Based Access
- ✓ Centralized Project & Resource Control
- ✓ Real-time Inventory & Booking
- ✓ Workflow Driven Processes
- ✓ Notifications & Alerts
- ✓ Document & Knowledge Repository
- ✓ Advanced Search & Analytics
- ✓ Scalable & Modular Architecture
- ✓ Audit Trail, Logs, Reports & Analytics

INTEGRATIONS

- ✉ Email / Communication
- 📱 SMS Gateway
- 💳 Payment Gateway (if any)
- 📅 Calendar / External Services

TECHNOLOGY STACK

Frontend : React / Angular
Backend : Spring Boot / Node.js
Database : MySQL / PostgreSQL
Cloud : AWS / Azure / GCP

BENEFITS

- ✓ Streamlined Project & Resource Management
- ✓ Better Utilization & Optimization
- ✓ Real-time Visibility & Tracking
- ✓ Improved Collaboration & Communication
- ✓ Data Driven Decisions & Reporting
- ✓ Enhanced Compliance & Accountability